
Contract Risk Review

Data Visualization Project

Project Details

This Contract Risk Review project transforms a JSON-based subcontract review into a four-page Power BI report. It organizes 25 clause findings into a structured view of contractual obligations, risks, opportunities, commercial terms, and schedule requirements.

The report covers a CAD 4.85 million mechanical subcontract, highlighting six high-severity findings, conditional delay damages, payment requirements, change authorization procedures, and notice periods. Clause references and source recommendations are retained so users can trace each finding back to the supplied review.

The deliverable includes four data tables, 12 DAX measures, and 26 visual elements. The semantic model was loaded successfully through Microsoft's Power BI MCP server, and 36 project files passed JSON schema validation. Visual definitions were subsequently adjusted for the older schema after a compatibility error in Power BI Desktop. Final rendering and data refresh remain unverified.

Business Objective

Help project and commercial teams prioritize contract findings, understand financial conditions, and track notice requirements without confusing potential exposure with actual losses.

Tools Used

Power BI, Power Query, DAX, JSON, Microsoft Power BI MCP Server, Node.js, and Python for schema validation.

Business Metrics in Focus

- **Total findings:** 25
- **High-severity findings:** 6, representing 24% of the review
- **Finding types:** 18 obligations, 4 risks, and 3 opportunities
- **Base contract value:** CAD 4,850,000, excluding applicable taxes
- **Conditional liquidated damages:** Up to CAD 15,000 per calendar day
- **Statutory holdback:** 10%, where required

Metric Mapping

Raw Column	Business Metric
clause_id	Finding count and clause-level traceability
severity	Severity distribution and high-severity share
finding_type	Obligations, risks, and opportunities
category	Distribution across Payment, Schedule, Delay, Change, and Other
financial_impact	Contract value, conditional damages, holdback, and markup limits extracted from narrative
deadline_days + supporting narrative	Notice periods, payment timing, and monthly submission cutoff
schedule_impact	Milestone dates and schedule conditions
confidence	Average source-reported confidence

Supporting Metrics

- **Severity distribution:** 6 High, 18 Medium, and 1 Low.
- **Average source confidence:** 95.36%; this is not a probability of financial loss.
- **Timing requirements:** Eight findings contain a populated deadline_days value.
- **Schedule milestones:** Seven dates are listed under clause 5.2.
- **Change markup limits:** 10% for self-performed work and 5% for lower-tier subcontracted work, unless otherwise agreed.
- **Payment timing:** Approved amounts are payable within 35 calendar days after approval of a complete and accurate application, subject to permitted withholding.

Columns from Data

Source fields	Source fields
clause_id	recommended_action
category	deadline_days
finding_type	financial_impact
severity	schedule_impact
finding	confidence

The source was organized into **Findings**, **Deadlines**, **Milestones**, and **Commercial terms** tables. Power Query embeds the supplied JSON snapshot, removing dependence on the original Downloads file.

Timing values were interpreted using their narrative context. For example, clause 4.1 requires submission by the **20th of each month**, rather than within a 20-day period. Business-day and calendar-day requirements remain distinct, and milestones are not marked overdue without completion information.

Dashboard Design Phases

Phase	Selection	Visual Type	Visual Size
Overview KPI Row	Total findings, high severity, high-severity share, source confidence	Four cards	Medium
Overview Analysis	Findings by severity, category, and finding type	Three horizontal bar charts	Medium
Priority Review	Six high-severity clauses and interpretation notes	Text panels	Large
Commercial KPI Row	Contract value, maximum daily damages, holdback rate	Three cards	Medium
Commercial Detail	Amounts, rates, clause references, and qualifying conditions	Table and explanatory text	Large
Notices and Milestones	Notice periods, triggers, payment administration, and milestone dates	Two tables and interpretation notes	Large
Findings Register	Complete source findings, recommendations, and impacts	Table with severity, category, and finding-type slicers	Full-width

DAX Measures

The project includes measures for finding counts, severity concentration, finding types, and source confidence. Contract amounts and rates are fixed reference measures extracted from the supplied review; they do not represent actual losses, retained balances, or accrued damages.

```
Total findings = COUNTROWS(Findings)
```

```
High severity =  
CALCULATE(  
    [Total findings],  
    Findings[severity] = "High"  
)
```

```
High share = DIVIDE([High severity], [Total findings])
```

```
Obligations =  
CALCULATE(  
    [Total findings],  
    Findings[finding_type] = "Obligation"  
)
```

```
Risks =  
CALCULATE(  
    [Total findings],  
    Findings[finding_type] = "Risk"  
)
```

```
Opportunities =  
CALCULATE(  
    [Total findings],  
    Findings[finding_type] = "Opportunity"  
)
```

```
Source confidence = AVERAGE(Findings[confidence])
```

```
Contract value CAD = 4850000
```

```
Maximum daily LD CAD = 15000
```

```
Holdback rate = 0.10
```

```
Self-performed markup cap = 0.10
```

```
Lower-tier markup cap = 0.05
```

Contract Risk Analyzer V2

Product Overview

Contract Risk Analyzer V2 is a Python-based product for extracting contract clauses, generating structured AI findings, and preparing data for Power BI reporting. The implementation is documented in **text-parser.ipynb**, the notebook used to produce the JSON for the Power BI reporting workflow.

Python Notebook and JSON Preparation

Extract and parse. `extract_pdf_pages()` uses PyMuPDF to read text page by page. `clean_text()` normalizes whitespace, and `parse_clauses()` uses regular expressions to identify sections and clauses. Records retain clause IDs, titles, text, section details, and start/end pages.

Add traceability and check source records. The code attaches document ID, document version, document type, project ID, and effective date to each clause. `validate_clauses()` checks for duplicate IDs, suspiciously short text, and invalid page ranges. Pandas exports the source records to `parsed_contract_clauses.csv`.

Analyze one clause at a time. `build_prompt()` supplies each clause and its source references to the LLM. A loop calls `client.responses.parse()` with the Pydantic `ClauseAnalysis` schema, defining category, finding type, severity, finding, recommended action, deadlines, impacts, and confidence.

Build and export the JSON. Each parsed response is converted to a dictionary with `model_dump()`, assigned the source `clause_id`, and appended to `results`. The notebook then writes the structured analysis records for downstream database and Power BI use:

```
with open("contract_risk_review(v2).json", "w") as f:
    json.dump(results, f, indent=4)
```

Governance and Project Status

Source clause text and tracking metadata remain in the clause dataset/CSV; the exported JSON contains AI analysis fields linked by `clause_id`. This separation distinguishes contract evidence from model interpretation. Pydantic validates the response structure, not the accuracy of the findings.

Contract Risk Analyzer V2 is currently a prototype; production deployment is not claimed. Preparing JSON for database and Power BI use does not establish production integration. No unverified quantitative performance results are claimed.

Implementation reference: `text-parser.ipynb`
/Users/faizaanm/Desktop/Contract Review Project/text-parser.ipynb